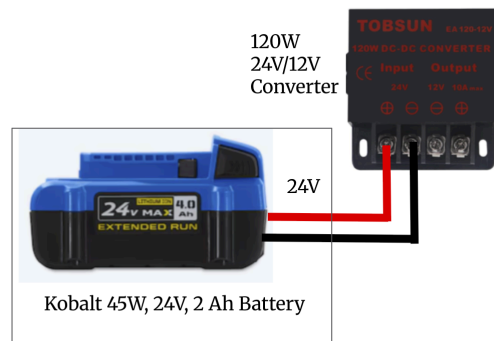


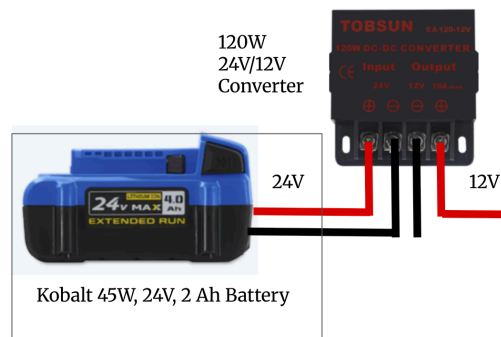
This section covers the build guide for connecting all electrical components and completing the circuit:

Sampling Unit Circuit:

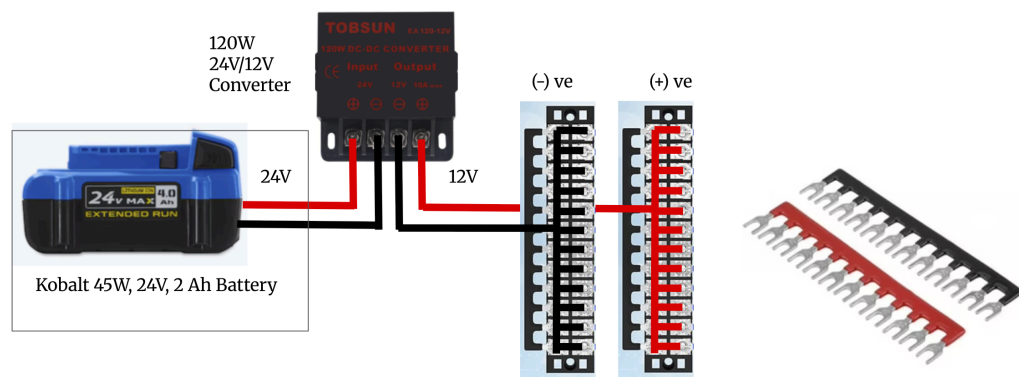
1. **Preparing Power Source:** This unit is powered by Kobalt 24V 2Ah battery which is stored in control unit housing and connected to a switch.
 - a. A 24V to 12V DC-DC step down converter is connected in 24V side using positive (Red) and negative (Black/Blue) wires from the battery.



- b. Connect another pair of positive (red) and black (negative) wires at the 12V side of the converter. Now that pair will work as the 12V supply for your system.



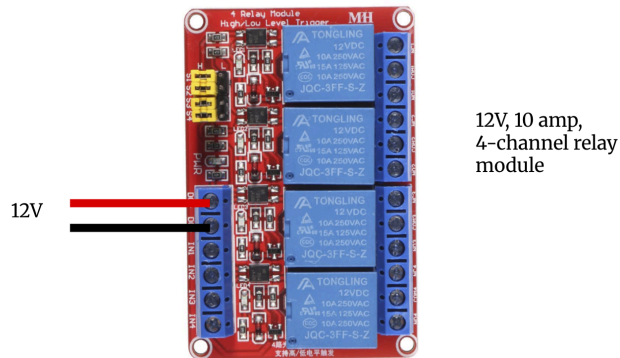
- c. Connect this 12V wire pair to two terminal blocks, and connect a rail (right fig) with each. In that way, one terminal block will be fully dedicated to the positive terminal and another to the negative terminal.



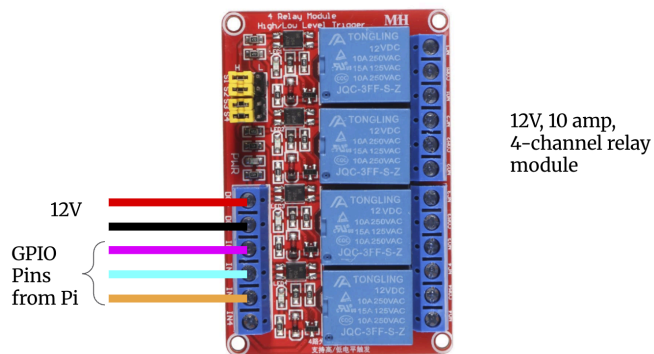
- d. You are all set to use the power system for the sampling unit.

2. Power and Trigger the 12V Relay Module:

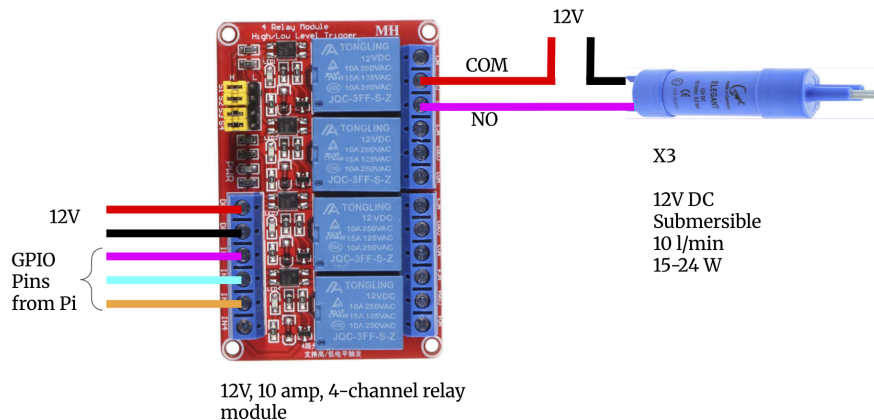
- Connect the positive (red) and negative (black) wires from 12V terminal blocks to DC+ and DC- terminals of the relay module.



- Connect the three GPIO pins for submersible pumps control, coming from the three Raspberry Pi GPIO pins in IN1-IN3 of the relay module. Now these IN1-IN3 will control relay 1-3 respectively.

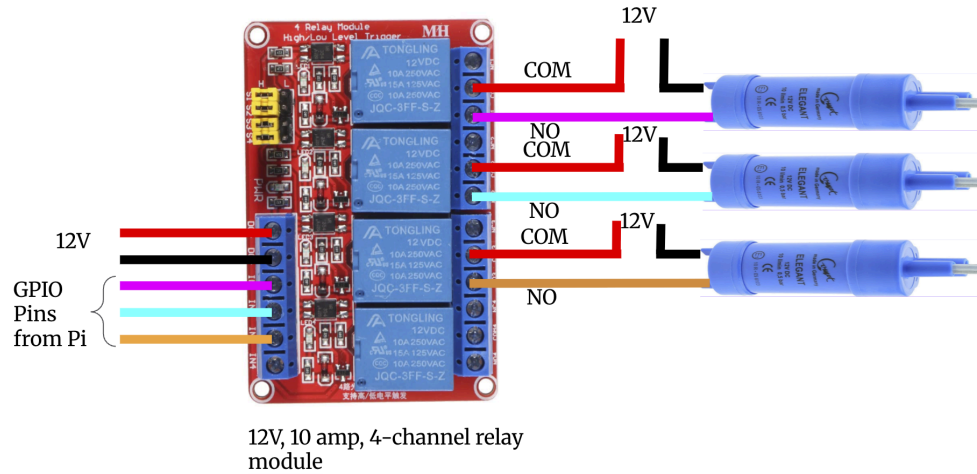


- Let's connect the 1st submersible pump. Connect one of the points of the positive terminal block to the COM1 port of relay 1 using a red wire. Then connect another wire from NO1 with a matching color as you used for IN1 (here, magenta), to the positive terminal of the pump.

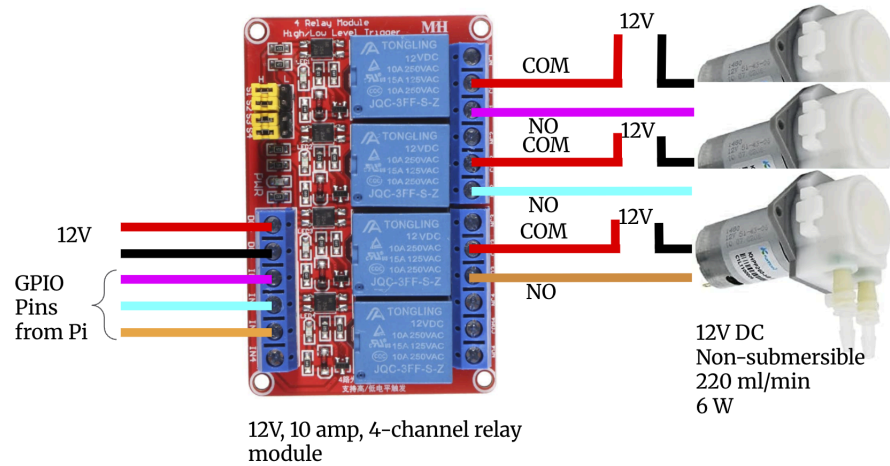


Connect the negative terminal block to the negative terminal of the pump using a black wire.

d. Repeat step c to connect all three submersible pumps.



e. Repeat steps a-d for another relay module to connect preservative pumps.

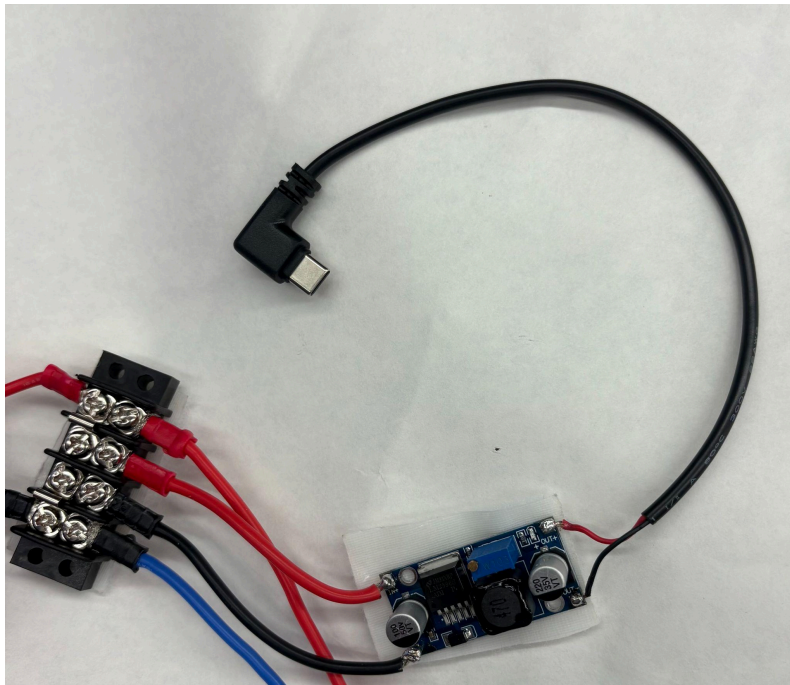


Tips:

1. Use 16 AWG wires to connect the components; they have very low internal resistance.
2. Use ferrules for connecting the wires to the terminal block and DC-DC converter.
3. Use the connectors wherever you need to connect/disconnect any wires frequently (like the pump's cables and the wires connected to the COM and NO of the relays)

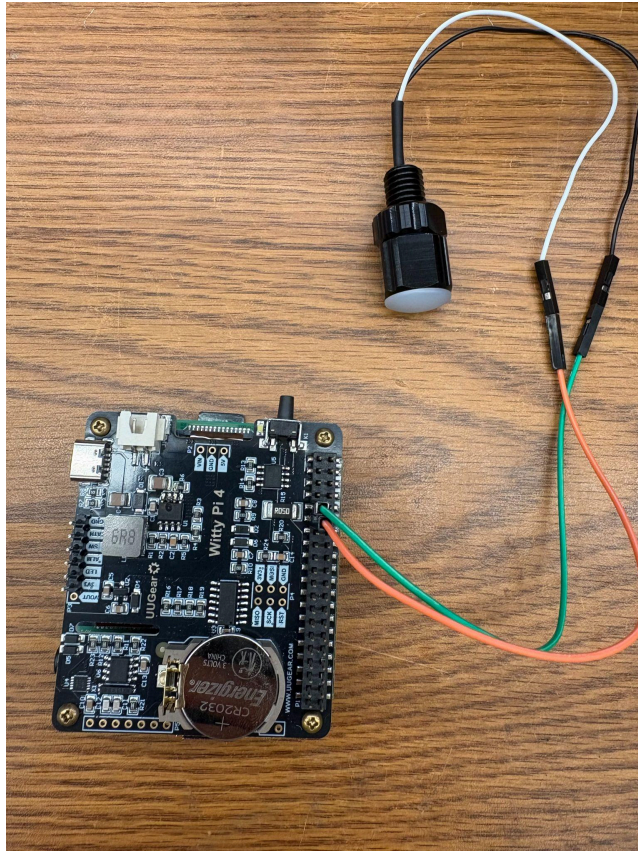
Control Unit Circuit:

1. **Connecting the Battery to the Witty Pi 4:** You will need to connect the battery's positive wire to a switch in series and connect the other end of the switch to a terminal block that will split the power supply into two units- one unit for the control unit and another unit for the sampling unit. Use connectors to connect only two consecutive rows together. In that way there will be a positive row pair and a negative row pair. Then you can connect two ferrules (one black (GND), one red (+V)), with two soldered 16 AWG wires to a Step Down converter, which should go from 24 V input to 5 V output. Then you will need to connect a usb-C male cable (and solder the two wires) to the output of the step down. When finished, it should look something like this.



NOTE: The red wire on the left connects to the positive battery end, and the black wire connects to the negative battery end. The blue and other red wire that go off the bottom of the image are routed to the Sampling Unit

2. **PINOUTS:** The first pins to connect are the GPIO pin and Ground pin used for the LED. The GPIO pin is pin 11 (The white/orange wire) and the GND pin is pin 9 (the black/green wire). Pin numbers correspond as the pinout diagram below the image shows.

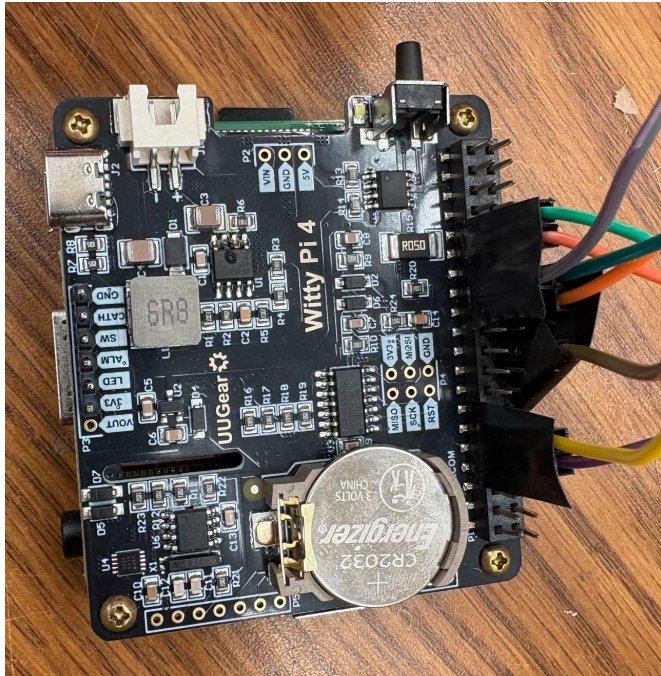


GPIO#	NAME		NAME	GPIO#
	3.3 VDC Power	1	5.0 VDC Power	2
8	GPIO 8 SDA1 (I2C)	3	5.0 VDC Power	4
9	GPIO 9 SCL1 (I2C)	5	Ground	6
7	GPIO 7 GPCLK0	7	GPIO 15 Tx0 (UART)	15
	Ground	9	GPIO 16 Rx0 (UART)	16
0	GPIO 0	11	GPIO 1 PCM_CLK/PWM0	1
2	GPIO 2	13	Ground	
3	GPIO 3	15	GPIO 4	4
	3.3 VDC Power	17	GPIO 5	5
12	GPIO 12 MOSI (SPI)	19	Ground	
13	GPIO 13 MISO (SPI)	21	GPIO 6	6
14	GPIO 14 SCLK (SPI)	23	GPIO 10 CE0 (SPI)	10
	Ground	25	GPIO 11 CE1 (SPI)	11
30	SDA0 (I2C ID EEPROM)	27	SCL0 (I2C ID EEPROM)	31
21	GPIO 21 GPCLK1	29	Ground	
22	GPIO 22 GPCLK2	31	GPIO 26 PWM0	26
23	GPIO 23 PWM1	33	Ground	
24	GPIO 24 PCM_FS/PWM1	35	GPIO 27	27
25	GPIO 25	37	GPIO 28 PCM_DIN	28
	Ground	39	GPIO 29 PCM_DOUT	29

Attention! The GPIO pin numbering used in this diagram is intended for use with WiringPi / Pi4J. This pin numbering is not the raw Broadcom GPIO pin numbers.

<http://www.pi4j.com>

- Regarding motor connections, the pins can be changed as desired (within [motor.py](#), and on the physical board) as long as none of them use pin 7, which is crucial to leave undisturbed. Currently [motor.py](#) is configured to be Main 1, 2, and 3 (pins 15, 16, 18), and Preservative 1, 2, and 3 (pins 22, 29, 31) . This is shown below, but is messy so be sure to use the pinout diagram to ensure proper connections.



4. If you ever want to have a button that operates as the black witty Pi button does now (for example you want a button that reaches outside the control unit), you can connect two wires, one to SW, and another to GND, and that will operate the exact same as hitting the button on the Witty Pi.